

$$1. L_1: \frac{x-1}{1} = \frac{y-2}{0} = \frac{z-3}{-1}, L_2: \frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{1}.$$

Find the plane passing through L_1 and parallel to L_2 .

Solution:

$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{vmatrix} = \mathbf{i} - 3\mathbf{j} + \mathbf{k}, \quad \text{and passing through } (1,2,3)$$

So the plane is $x - 3y + z + 2 = 0$

$$2. \text{ Does the series } \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \text{ converge or diverge?}$$

Solution:

$$p > 1, \exists q (p > q > 1), \frac{\ln^2 n}{n^p} \leq \frac{1}{n^q}, \sum_{n=2}^{\infty} \frac{1}{n^q} \text{ converge} \Rightarrow \sum_{n=2}^{\infty} \frac{\ln^2 n}{n^p} \text{ converge}$$

$$\text{So } \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \text{ converge absolutely}$$

$$p \leq 1, \frac{\ln^2 n}{n^p} \geq \frac{\ln^2 n}{n} \geq \frac{1}{n}, \sum_{n=2}^{\infty} \frac{1}{n} \text{ diverge} \Rightarrow \sum_{n=2}^{\infty} \frac{\ln^2 n}{n^p} \text{ diverge}$$

$$0 < p < 1, \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \text{ converge} \Rightarrow \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \text{ conditional convergence}$$

$$p \leq 0, \lim_{n \rightarrow \infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \neq 0 \Rightarrow \sum_{n=2}^{\infty} (-1)^{n+1} \frac{\ln^2 n}{n^p} \text{ diverge}$$

$$3. \text{ Find convergence set and sum for the power series } \sum_{n=1}^{\infty} (-1)^{n-1} \left(1 + \frac{1}{n(2n-1)}\right) x^{2n}.$$

Solution:

$$A = \lim_{n \rightarrow \infty} \sqrt{\left| (-1)^{n-1} \left(1 + \frac{1}{n(2n-1)}\right) \right|} = 1, R = \frac{1}{A} = 1, \sum_{n=1}^{\infty} (-1)^{n-1} \left(1 + \frac{1}{n(2n-1)}\right) \text{ diverge}$$

So convergence set $D = (-1,1)$

$$\text{Let } S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n}, x \in (-1,1), \quad S(0) = 0$$

$$S'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)} x^{2n-1}, x \in (-1,1), \quad S'(0) = 0$$

$$S''(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} = \frac{1}{1+x^2}, x \in (-1,1)$$

$$S'(x) = \int_0^x S''(t) dt = \int_0^x \frac{1}{1+t^2} dt = \arctan x$$

$$S(x) = \int_0^x S'(t) dt = \int_0^x \arctan t dt = x \arctan x - \frac{1}{2} \ln(1+x^2)$$

$$\sum_{n=1}^{\infty} (-1)^{n-1} x^{2n} = \frac{x^2}{1+x^2}, x \in (-1,1)$$

$$\sum_{n=1}^{\infty} (-1)^{n-1} \left(1 + \frac{1}{n(2n-1)}\right) x^{2n} = 2S(x) + \frac{x^2}{1+x^2} = 2x \arctan x - \ln(1+x^2) + \frac{x^2}{1+x^2}$$

$$x \in (-1,1)$$

4. Expand the function $f(x) = \sum_{n=1}^{\infty} \left(\frac{x}{1-x}\right)^n$ into power series of x .

Solution:

$$\left|\frac{x}{1-x}\right| < 1 \Rightarrow |x| < \frac{1}{2}, f(x) = \sum_{n=1}^{\infty} \left(\frac{x}{1-x}\right)^n = \frac{x}{1-2x}$$

$$\frac{x}{1-2x} = \frac{1}{2} \cdot \frac{2x}{1-2x} = \frac{1}{2} \sum_{n=1}^{\infty} (2x)^n = \sum_{n=1}^{\infty} 2^{n-1} x^n \cdot \left(|x| < \frac{1}{2}\right)$$

5. Determine the continuity of $f(x,y) = \begin{cases} x^2 \sin \frac{1}{x^2+y^2} + y^2 & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

Solution:

$$\lim_{\substack{y \rightarrow 0 \\ x \rightarrow 0}} \frac{x^2 \sin \frac{1}{x^2+y^2} + y^2}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x^2}}{x^2} = \lim_{x \rightarrow 0} \sin \frac{1}{x^2}$$

$$\lim_{x \rightarrow 0} \sin \frac{1}{x^2} \text{ not exist} \Rightarrow f(x,y) \text{ is discontinuous at } (0,0)$$

$$6. f(x,y) = \begin{cases} xy \frac{x^2-y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}, \quad \text{find } f_{xy}(0,0), f_{yx}(0,0).$$

Solution:

$$(x,y) \neq (0,0), f_x(x,y) = \frac{y(x^4 - y^4 + 4x^2y^2)}{(x^2+y^2)^2}, f_y(x,y) = \frac{y(x^4 - y^4 - 4x^2y^2)}{(x^2+y^2)^2}$$

$$(x,y) = (0,0), f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x} = 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y} = 0$$

$$f_{xy}(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f_x(0,0+\Delta y) - f_x(0,0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\frac{\Delta y(-\Delta y^4)}{\Delta y^4} - 0}{\Delta y} = -1$$

$$f_{yx}(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f_y(0 + \Delta x, 0) - f_y(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x \cdot \Delta x^4}{\Delta x^4} - 0}{\Delta x} = 1$$

7. Let $z = f\left(xy, \frac{x}{y}\right) + g\left(\frac{y}{x}\right)$, f has a second – order continuous partial derivative,

g has a second – order continuous derivative, find $\frac{\partial^2 z}{\partial x \partial y}$.

Solution:

$$\frac{\partial z}{\partial x} = y f_1 + \frac{1}{y} f_2 - \frac{y}{x^2} g'$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= f_1 + y \left(x f_{11} - \frac{x}{y^2} f_{12} \right) - \frac{1}{y^2} f_2 + \frac{1}{y} \left(x f_{12} - \frac{x}{y^2} f_{22} \right) - \frac{1}{x^2} g' - \frac{y}{x^3} g'' \\ &= f_1 - \frac{1}{y^2} f_2 + x y f_{11} - \frac{x}{y^3} f_{22} - \frac{1}{x^2} g' - \frac{y}{x^3} g'' \end{aligned}$$

8. Find the tangent line to $\begin{cases} x^2 + y^2 + z^2 = 6 \\ z = x^2 + y^2 \end{cases}$ at $(-1, 1, 2)$.

Solution:

$$\begin{cases} F(x, y, z) = x^2 + y^2 + z^2 - 6 = 0 \\ G(x, y, z) = z - x^2 - y^2 = 0 \end{cases}$$

$$\tau = \left(\frac{\partial(F, G)}{\partial(y, z)}, \frac{\partial(F, G)}{\partial(z, x)}, \frac{\partial(F, G)}{\partial(x, y)} \right) = (2y(1 + 2z), -2x(1 + 2z), 0)$$

At $(-1, 1, 2)$, $\tau = (10, 10, 0)$

$$\text{So tangent line: } \frac{x + 1}{10} = \frac{y - 1}{10} = \frac{z - 2}{0}$$

9. Let $u = f(x, y, z)$ has a continuous partial derivative, $z = z(x, y)$ is determined by $x e^x - y e^y = z e^z$, find du .

Solution:

$$F(x, y, z) = x e^x - y e^y - z e^z, F_x = (x + 1)e^x, F_y = -(y + 1)e^y, F_z = -(z + 1)e^z$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{x + 1}{z + 1} e^{x-z}, \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{y + 1}{z + 1} e^{y-z}$$

$$\frac{\partial u}{\partial x} = f_x + f_z \frac{\partial z}{\partial x} = f_x + f_z \frac{x + 1}{z + 1} e^{x-z}, \frac{\partial u}{\partial y} = f_y + f_z \frac{\partial z}{\partial y} = f_y - f_z \frac{y + 1}{z + 1} e^{y-z}$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = \left(f_x + f_z \frac{x + 1}{z + 1} e^{x-z} \right) dx + \left(f_y - f_z \frac{y + 1}{z + 1} e^{y-z} \right) dy$$

10. Find the maximum and minimum values of $z = f(x, y) = x^2 y(4 - x - y)$ on the region D , $D = \{(x, y) | x \geq 0, y \geq 0, x + y \leq 6\}$.

Solution:

$$D^o = \{(x, y) | x > 0, y > 0, x + y < 6\}$$

$$\begin{cases} f_x(x, y) = 2xy(4 - x - y) - x^2 y = 0 \\ f_y(x, y) = x^2(4 - x - y) - x^2 y = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$$

$$f_{xx} = 8y - 6xy - 2y^2, f_{xy} = 8x - 3x^2 - 4xy, f_{yy} = -2x^2$$

$$A = f_{xx}(2, 1) = -6 < 0, B = f_{xy}(2, 1) = -4, C = f_{yy}(2, 1) = -8$$

$$AC - B^2 = 32 > 0 \Rightarrow (2, 1) \text{ is local maximum point, } f(2, 1) = 4$$

When $x = 0$ ($0 \leq y \leq 6$), $f(x, y) = 0$. When $y = 0$ ($0 \leq x \leq 6$), $f(x, y) = 0$

When $x + y = 6, y = 6 - x, z = f(x, 6 - x) = 2x^3 - 12x^2 (0 \leq x \leq 6)$

$$\Rightarrow z_{\min} = f(4, 2) = -64$$

$$\text{So } f_{\max} = f(2, 1) = 4, f_{\min} = f(4, 2) = -64$$

11. Find the minimum distance from $y = x^2$ to $x - y - 2 = 0$.

Solution:

$$\forall (x_1, y_1), (x_2, y_2). y_1^2 = x_1^2, (x_2, y_2), x_2 - y_2 - 2 = 0$$

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

$$\text{Let } F(x_1, x_2, y_1, y_2, \lambda_1, \lambda_2) = (x_2 - x_1)^2 + (y_2 - y_1)^2 + \lambda_1(y_1 - x_1^2) + \lambda_2(x_2 - y_2 - 2)$$

$$\begin{cases} \frac{\partial F}{\partial x_1} = -2(x_2 - x_1) - 2\lambda_1 x_1 = 0 \\ \frac{\partial F}{\partial x_2} = 2(x_2 - x_1) + \lambda_2 = 0 \\ \frac{\partial F}{\partial y_1} = -2(y_2 - y_1) + \lambda_1 = 0 \\ \frac{\partial F}{\partial y_2} = 2(y_2 - y_1) - \lambda_2 = 0 \\ y_1 = x_1^2 \\ x_2 - y_2 - 2 = 0 \end{cases}$$

$$\Rightarrow x_1 = \frac{1}{2}, y_1 = \frac{1}{4}, x_2 = \frac{11}{8}, y_2 = -\frac{5}{8}$$

$$\Rightarrow d_{\min} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \frac{7}{8}\sqrt{2}$$

12. Find $\int \int_D \frac{3x}{y^2 + xy^3} dx dy$, D is bounded by $xy = 1, xy = 3, y^2 = x$ and $y^2 = 3x$.

Solution:

$$\text{Let } \begin{cases} u = xy \\ v = \frac{y^2}{x} \end{cases}, J = \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\frac{\partial(u, v)}{\partial(x, y)}} = \frac{1}{3v}, D' = \{(u, v) | 1 \leq u \leq 3, 1 \leq v \leq 3\}$$

$$\int \int_D \frac{3x}{y^2 + xy^3} dx dy = \int \int_{D'} \frac{3}{v(1+u)} |J| du dv = \int_1^3 \frac{1}{v^2} dv \int_1^3 \frac{1}{1+u} du = \frac{2}{3} \ln 2$$

13. Compute the triple integral $\int \int \int_{\Omega} x^2 \sqrt{x^2 + y^2} dx dy dz$,

Ω is bounded by $z = \sqrt{x^2 + y^2}$ and $z = x^2 + y^2$.

Solution:

$$\text{Let } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}, J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r, \Omega' = \{(r, \theta, z) | 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, r^2 \leq z \leq r\}$$

$$\begin{aligned} \int \int \int_{\Omega} x^2 \sqrt{x^2 + y^2} dx dy dz &= \int \int \int_{\Omega'} r^2 \cos^2 \theta \cdot r \cdot |J| dr d\theta dz \\ &= \int_0^{2\pi} \cos^2 \theta d\theta \int_0^1 r^4 dr \int_{r^2}^r dz = \frac{\pi}{42} \end{aligned}$$

14. Let L be $y = \pi \cos x$ from $A(\pi, -\pi)$ to $B(-\pi, -\pi)$, find $\int_L \frac{(x+y)dx - (x-y)dy}{x^2 + y^2}$.

Solution:

Let $L_1: x^2 + y^2 = R^2 (R \ll 1)$, the direction is clockwise

$$P(x, y) = \frac{x+y}{x^2+y^2}, Q(x, y) = -\frac{x-y}{x^2+y^2}, \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

By Green Formula

$$\int_L \frac{(x+y)dx - (x-y)dy}{x^2+y^2} + \int_{L_1} \frac{(x+y)dx - (x-y)dy}{x^2+y^2} + \int_{\overline{BA}} \frac{(x+y)dx - (x-y)dy}{x^2+y^2} = 0$$

$$\int_{\overline{BA}} \frac{(x+y)dx - (x-y)dy}{x^2+y^2} = \int_{-\pi}^{\pi} \frac{x-\pi}{x^2+\pi^2} dx = -\frac{\pi}{2}$$

$$\begin{aligned} & \int_{L_1} \frac{(x+y)dx - (x-y)dy}{x^2+y^2} \\ &= \int_{2\pi}^0 \frac{(R \cos t + R \sin t)(-R \sin t) - (R \cos t - R \sin t)R \cos t}{R^2} dt = 2\pi \end{aligned}$$

$$\text{So } \int_L \frac{(x+y)dx - (x-y)dy}{x^2+y^2} = -\frac{3}{2}\pi$$

$$15. \text{ Find } I = \int_{\Sigma} \frac{1}{\sqrt{x^2+y^2+(z-a)^2}} dS, \quad \Sigma: x^2+y^2+z^2=R^2, 0 \leq a < +\infty, a \neq R.$$

Solution:

$$\text{Let } \begin{cases} x = R \sin \varphi \cos \theta \\ y = R \sin \varphi \sin \theta, \theta \in [0, 2\pi], \varphi \in [0, \pi] \\ z = R \cos \varphi \end{cases}$$

$$\begin{cases} E = x_{\varphi}^2 + y_{\varphi}^2 + z_{\varphi}^2 = R^2 \\ G = x_{\theta}^2 + y_{\theta}^2 + z_{\theta}^2 = R^2 \sin^2 \varphi \\ F = x_{\varphi} x_{\theta} + y_{\varphi} y_{\theta} + z_{\varphi} z_{\theta} = 0 \end{cases}$$

$$\begin{aligned} I &= \int \int_{\Sigma} \frac{1}{\sqrt{x^2+y^2+(z-a)^2}} dS = \int_0^{2\pi} d\theta \int_0^{\pi} \frac{1}{\sqrt{R^2+a^2-2aR \cos \varphi}} R^2 \sin \varphi d\varphi \\ &= 2\pi R^2 \left[\frac{1}{aR} \sqrt{R^2+a^2-2aR \cos \varphi} \right] \Big|_0^{\pi} = \begin{cases} 4\pi R, & R > a \\ \frac{4\pi R^2}{a}, & R < a \end{cases} \end{aligned}$$

$$16. \text{ Find } \int \int_{\Sigma} \frac{axdydz + (z+a)^2 dxdy}{\sqrt{x^2+y^2+z^2}}, \quad \Sigma: z = -\sqrt{a^2-x^2-y^2}, \quad (a > 0)$$

the direction is upper side.

Solution:

$$\text{Let } \Sigma_1: \begin{cases} x^2+y^2 \leq a^2 \\ z=0 \end{cases}, \text{ the direction is lower side}$$

$$\Omega = \{(x, y, z) | x^2+y^2+z^2 \leq a^2, z \leq 0\}$$

$$I = \int \int_{\Sigma} \frac{axdydz + (z+a)^2 dxdy}{\sqrt{x^2+y^2+z^2}} = \frac{1}{a} \int \int_{\Sigma} axdydz + (z+a)^2 dxdy$$

By Gauss Formula

$$\int \int_{\Sigma+\Sigma_1} axdydz + (z+a)^2 dxdy = - \int \int \int_{\Omega} (3a+2z) dxdydz$$

$$\int \int \int_{\Omega} (3a+2z) dxdydz = \int \int \int_{\Omega} 3a dxdydz + \int \int \int_{\Omega} 2z dxdydz$$

$$\int \int \int_{\Omega} 3a dxdydz = 3a \cdot \frac{4}{3} \pi a^3 \cdot \frac{1}{2} = 2\pi a^4$$

$$\int \int \int_{\Omega} 2z \, dx dy dz = 2 \int_0^{2\pi} d\theta \int_{\frac{\pi}{2}}^{\pi} d\varphi \int_0^a r \cos \varphi \cdot r^2 \sin \varphi \, dr = -\frac{1}{2} \pi a^4$$

$$\Rightarrow \int \int \int_{\Omega} (3a + 2z) \, dx dy dz = \frac{3}{2} \pi a^4$$

$$\int \int_{\Sigma_1} ax dy dz + (z+a)^2 dx dy = - \int \int_D a^2 dx dy = -\pi a^4$$

$$\text{So } I = \int \int_{\Sigma} \frac{ax dy dz + (z+a)^2 dx dy}{\sqrt{x^2 + y^2 + z^2}} = \frac{1}{a} \cdot \left(-\frac{3}{2} \pi a^4 + \pi a^4 \right) = -\frac{\pi}{2} a^3$$

17. Find the solution of the equation $(2x + y - 4)dx + (x + y - 1)dy = 0$.

Solution:

$$\frac{dy}{dx} = -\frac{2x + y - 4}{x + y - 1}, \begin{cases} 2x + y - 4 = 0 \\ x + y - 1 = 0 \end{cases} \Rightarrow \begin{cases} x = 3 \\ y = -2 \end{cases}$$

$$\text{Let } \begin{cases} X = x - 3 \\ Y = y + 2 \end{cases} \Rightarrow \frac{dY}{dX} = -\frac{2X + Y}{X + Y} = -\frac{2 + \frac{Y}{X}}{1 + \frac{Y}{X}}, \text{ Let } u = \frac{Y}{X}, Y = uX$$

$$\Rightarrow \frac{dY}{dX} = X \frac{du}{dX} + u \Rightarrow X \frac{du}{dX} + u = -\frac{2+u}{1+u} \Rightarrow -\frac{u+1}{u^2+2u+2} du = \frac{dX}{X}$$

$$\Rightarrow \ln C_1 - \frac{1}{2} \ln(u^2 + 2u + 2) = \ln X \Rightarrow \frac{C_1}{\sqrt{u^2 + 2u + 2}} = X,$$

$$\Rightarrow Y^2 + 2XY + 2X^2 = C_2 (C_2 = C_1^2)$$

$\Rightarrow$ The solution of the equation is $2x^2 + 2xy + y^2 - 8x - 2y = C (C = C_2 - 10)$

18. Find the solution of the equation $y'' - 2y' + 2y = e^x \cos x + e^{2x}$.

Solution:

Consider homogeneous linear differential equation $y'' - 2y' + 2y = 0$

characteristic equation $r^2 - 2r + 2 = 0$

characteristic root $r_1 = 1 + i, r_2 = 1 - i$

$\Rightarrow$ General solutions of homogeneous equation are $y = C_1 e^x \cos x + C_2 e^x \sin x$

Consider equation $y'' - 2y' + 2y = e^x \cos x$

$r = 1 + i$ is the single root of the characteristic equation

So special solution $\widetilde{y}_1 = x(A \cos x + B \sin x)e^x$

Substitute into $y'' - 2y' + 2y = e^x \cos x \Rightarrow A = 0, B = \frac{1}{2}$

$\Rightarrow$ special solution $\widetilde{y}_1 = \frac{1}{2} x \sin x e^x$

Consider equation $y'' - 2y' + 2y = e^{2x}$

$r = 2$ is not the root of the characteristic equation

So special solution $\widetilde{y}_2 = C e^{2x}$

Substitute into $y'' - 2y' + 2y = e^{2x} \Rightarrow C = \frac{1}{2}$

$\Rightarrow$ special solution $\widetilde{y}_2 = \frac{1}{2} e^{2x}$

$\Rightarrow$ General solutions of nonhomogeneous equation are

$$y = C_1 e^x \cos x + C_2 e^x \sin x + \frac{1}{2} x \sin x e^x + \frac{1}{2} e^x$$